BAHRIA UNIVERSITY, ISLAMABAD

Department of Computer Science

INTRODUCTION TO DATA SCIENCE LAB

LAB JOURNAL # 07

SUBMITTED BY

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6TH A

SUBMITTED TO

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1 TASK 1:

Perform the following preprocessing techniques on the reviews of movies data:

- Lowercasing
- Remove HTML
- Remove URLs
- Remove punctautions
- Stop words removal
- Tokenization
- Tf
- Tf/idf

2 SOLUTION

```
import pandas as pd
import re
import string
import math
# Load dataset
df = pd.read_csv("Week7\\IMDB Dataset.csv")
# Punctuation and stop words
table = str.maketrans("", "", string.punctuation)
stop =
{"the","is","a","an","and","of","to","in","this","was","it","i","very","are","
on"}
# 1. Lowercasing
# 2. Remove HTML
# 3. Remove URLs
# 4. Remove punctuation
# 5. Stop words removal
# 6. Tokenization
def preprocess(text):
    text = text.lower()
    text = re.sub(r"<.*?>", "", text)
    text = re.sub(r"https?://\S+|www\.\S+", "", text)
    text = text.translate(table)
    return [w for w in re.findall(r"\w+", text) if w not in stop]
tok = df["review"].apply(preprocess).tolist()
print(tok[0])
# 7. Term Frequency (TF)
# 8. TF-IDF
tf  = lambda t, d: d.count(t) / len(d)
idf = lambda t: math.log(len(tok) / sum(t in d for d in tok))
```

```
for t in ["movie", "film", "good", "bad"]:
    print(t, "tf =", round(tf(t, tok[1]), 2), "idf =", round(idf(t), 2),
          "tf-idf =", round(tf(t, tok[1]) * idf(t), 2))
```

3 OUTPUT

```
['one', 'other', 'reviewers', 'has', 'mentioned', 'that', 'after', 'watching', 'just', '1', 'oz', 'episode', 'youll', 'be', 'hooked', 'they', 'right', 'as', 'exactly', 'what', 'happened', 'with', 'methe', 'first', 'thing', 'that', 'struck', 'me', 'about', 'oz', 'its', 'brutality', 'unflinching', 'scenes', 'violence', 'which', 'set', 'right', 'from', 'word', 'go', 'trust', 'me', 'not', 'show', 'for', 'faint', 'hearted', 'or', 'timid', 'show', 'pulls', 'no', 'punches', 'with', 'regards', 'drugs', 'sex', 'or', 'violence', 'its', 'hardcore', 'classic', 'use', 'wordit', 'called', 'oz', 'as', 'that', 'nickname', 'given', 'oswald', 'maximum', 'security', 'state', 'penitentary', 'focuses', 'mainly', 'emerald', 'city', 'experimental', 'section', 'prison', 'where', 'all', 'cells', 'have', 'glass', 'fronts', 'face', 'inwards', 'so', 'privacy', 'not', 'high', 'agenda', 'em', 'city', 'home', 'manyaryans', 'muslims', 'gangstas', 'latinos', 'christians', 'italians', 'irish', 'moreso', 'scuffles', 'death', 'stares', 'dodgy', 'dealings', 'shady', 'agreements', 'never', 'far', 'awayi', 'would', 'say', 'main', 'appeal', 'show', 'due', 'fact', 'that', 'goes', 'where', 'other', 'shows', 'wouldnt', 'dare', 'forget', 'pretty', 'pictures', 'painted', 'for', 'mainstream', 'audiences', 'forget', 'charm', 'forget', 'romanceoz', 'doesnt', 'mess', 'around', 'first', 'episode', 'ever', 'saw', 'struck', 'me', 'as', 'so', 'nasty', 'surreal', 'couldnt', 'say', 'ready', 'for', 'but', 'as', 'watched', 'more', 'developed', 'taste', 'for', 'oz', 'got', 'accustomed', 'high', 'levels', 'graphic', 'violence', 'not', 'just', 'violence', 'but', 'injustice', 'crooked', 'guards', 'wholl', 'be', 'sold', 'out', 'for', 'nickel', 'inmates', 'wholl', 'kill', 'order', 'get', 'away', 'with', 'well', 'mannered', 'middle', 'class', 'inmates', 'being', 'turned', 'into', 'prison', 'bitches', 'due', 'their', 'lack', 'street', 'skills', 'or', 'prison', 'experience', 'watching', 'oz', 'you', 'may', 'become', 'comfortable', 'with', 'what', 'uncomfortable', 'viewingthats', 'if', 'you', 'can', 'get', 'touch', 'with', 'your', 'darker', 'side']
movie tf = 0.0 idf = 0.51 tf-idf = 0.0
film tf = 0.0 idf = 0.61 tf-idf = 0.0
good tf = 0.0 idf = 1.0 tf-idf = 0.0
bad tf = 0.0 idf = 1.48 tf-idf = 0.0
```